

1 Grammar

```
<program> ::= <def-list> <term> | <term>

<def-list> ::= <def> | <def> <new-strings> <def-list>

<def> ::= "let " <var> " =" <term>

<new-strings> ::= "\\n" | "\\n" <new-strings>

<term> ::= <term'> | <term'> <spaces> <term>

<term'> ::= "\" <var-list> "." <term> | <var> | "(" <term> ")"

<var-list> ::= <var> | <var> <spaces> <var-list>

<var> ::= a-z | A-Z , { a-z | A-Z | 0-9 | "_" }

<spaces> ::= " " | " " <spaces>
```

2 Derivation

2.1 \x.x

```
<program> => <term> => <term'> => \<var-list>.<term> => \<var>.<term> => \x.<term> =>
=> \x.<term'> => \x.<var> => \x.x
```

2.2 (\x.x) y

```
<program> => <term> => <term'> <spaces> <term> => (<term>) <spaces> <term> =>
=> (<term'>) <spaces> <term> => (\<var-list>.<term>) <spaces> <term> =>
=> (\<var>.<term>) <spaces> <term> => (\x.<term>) <spaces> <term> =>
=> (\x.<term'>) <spaces> <term> => (\x.<var>) <spaces> <term> => (\x.x) <spaces> <term> =>
=> (\x.x) <term> => (\x.x) <term'> => (\x.x) <var> => (\x.x) y
```